

Pulkit Dubey

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Summary

PhD in Applied Mathematics with expertise in computational mathematics, physics simulations, and numerical optimization. Experienced in developing high-performance software frameworks in Python, Julia and C++, with working knowledge of MPI, CUDA, and PyTorch for scalable parallel computing over HPC clusters. Proven expertise in integrating neural networks with physics based solvers. Strong communicator with collaborative publications and presentations at major scientific computing conferences.

Experience

University of California, Santa Cruz

Santa Cruz, CA

Postdoctoral Researcher

Oct 2025 - Present

- **Optimal control of chaotic systems using differentiable neural emulators:** Developed gradient-based optimization pipelines in PyTorch and JAX, using state-of-the-art ML weather emulators (LUCIE, NeuralGCM, ACE2-ERA5) as differentiable surrogate models for atmospheric control. Implemented Backpropagation-Through-Time with gradient checkpointing to scale long-horizon optimization to HPC environments, and mitigate chaotic gradient artifacts via successive short-horizon ensemble averaging.
- **ML data infrastructure and emulator training for climate applications:** Built end-to-end pipelines for ERA5 reanalysis data, including regridding and quality validation of multi-variable atmospheric fields at global scale. Trained and fine-tuned an SFNO-based neural weather model on ERA5, extending its variable set and validating checkpoint quality against held-out fields; developed modular post-processing and visualization tooling across multiple emulator architectures.

University of New Hampshire

Durham, NH

Doctoral Researcher

Jan 2020 - Sept 2025

- **Reduced order modeling of turbulent flows:** Developed scalable Python simulation frameworks using spectral methods to handle high-dimensional turbulent flow data from large-scale physics simulations. Leveraged asymptotic analysis for reduced-order modeling, achieving a 10× performance improvement. Implemented custom data structures for efficient post-processing, management, and visualization of simulation statistics. Presented research at APS-DFD 2024; manuscript in preparation for the Journal of Fluid Mechanics.
- **Accelerating numerical solvers with neural networks:** Designed a C++/CUDA framework that interleaves a high-fidelity CFD solver with a PyTorch neural operator: the solver advances 3–5 timesteps, the network forecasts the next window, then returns control to physics. By capping the NN horizon at the chaos limit, the method halves wall-clock time while preserving accuracy and keeps the entire pipeline differentiable for downstream optimization or control tasks. The work was presented at the SC24 supercomputing conference in Atlanta, GA.
- **Teaching assistant for undergraduate courses:** Delivered classroom lectures and mentored students through undergraduate engineering courses. Developed an automated grading system streamlining faculty workload and accelerating student feedback. Recognized with competitive fellowships for outstanding TAs.

TIFR - Center for Applicable Mathematics

Bangalore, India

Intern - Airbus Chair for Mathematics of Complex Systems

Feb 2016 - June 2016

- **Development of open-source numerical solver in C:** Developed a finite-volume PDE solver for high speed flows in C for the Navier-Stokes equations. Maintained a Git-based workflow for version control, ensuring code portability and collaboration.

Tata Motors

Pune, India

Graduate Engineer Trainee

May 2015 - Jan 2016

- **Thermal system analyses in IC Engine:** FEM-based studies of crankshaft deflections under extreme loads in ANSYS and CATIA. Compared multiple vendor designs to optimize cost, safety, and structural integrity. Earned Siemens PLM certification.

Relevant Projects

CAD modeling of MEMS devices: Design of MEMS vibrational energy harvester and gas sensor in Solidworks and coupled thermo-mechanical analysis in ANSYS.

Effects of size of a neural network on overfitting: Approximating mathematical functions with neural networks of

varying sizes, starting from a single neuron up to deeper, multi-layer architectures to study overfitting, accuracy and training time. Optimized the training over a CUDA enabled desktop GPU. Gained hands-on experience with the end-to-end ML workflow including model development, training and evaluation.

Neural Surrogate for chaotic dynamical systems: MATLAB based simulation of Lorentz attractor that trains a neural surrogate; benchmarked data-driven predictions against ground truth from ODE solvers to study accuracy and generalization. Extendable to differentiable physics engines for real-time control.

Education

University of New Hampshire

PhD, Applied Mathematics (4.0/4.0)

Durham, NH

Current

Jawaharlal Nehru Centre for Advanced Scientific Research

M.S.(Engg.) Engineering Mechanics and Applied Mathematics

Bangalore, India

May 2019

BITS - Pilani

B.E.(Hons.) Mechanical Engineering

Pilani, India

August 2015

Recognition

Referee for the prestigious journal, Journal of Fluid Mechanics (2023)

Summer fellowship for outstanding TAs (2023,2024)

Fellow for the prestigious Boulder Summer School in Condensed Matter Physics (2022)

Selected for the Airbus Chair on Mathematics of Complex Systems (2016)

Conferences

APS-DFD 2022, 2023, 2024

Boulder School for Condensed Matter Physics 2022

SC24 (International conference on High Performance Computing)

Relevant Publications

Atif, Mohammad, Pulkit Dubey, et al. (2024). "Fourier neural operators for spatiotemporal dynamics in two-dimensional turbulence". In: *arXiv preprint arXiv:2409.14660*.

Dubey, Pulkit, Rama Govindarajan, and Greg Chini (2025). "Quasilinear simulations of viscosity stratified flows". In: *Under preparation for Journal of Fluid Mechanics*.

Dubey, Pulkit, Anubhab Roy, and Ganesh Subramanian (2022). "Linear stability of a rotating liquid column revisited". In: *Journal of Fluid Mechanics* 933, A55. doi: 10.1017/jfm.2021.1109.

Kaushal, Shaurya, Pulkit Dubey, et al. (2015). "Novel design for wideband piezoelectric vibrational energy harvester (P-VEH)". In: *2015 19th International Symposium on VLSI Design and Test. IEEE*, pp. 1–5.